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Abiraterone plus prednisone added to androgen deprivation therapy and docetaxel in de novo metastatic castration-sensitive prostate cancer (PEACE-1): a multicentre, open-label, randomised, phase 3 study with a 2 × 2 factorial design

[Prof Karim Fizazi, MD](#) ^a · [Stéphanie Foulon, MD](#) ^{b,c} · [Prof Joan Carles, MD](#) ^e · [Guilhem Roubaud, MD](#) ^f · [Prof Ray McDermott, MD](#) ^g · [Aude Fléchon, MD](#) ^h · [Prof Bertrand Tombal, MD](#) ⁱ · [Prof Stéphane Supiot, MD](#) ^j · [Dominik Berthold, MD](#) ^k · [Philippe Ronchin, MD](#) ^l · [Gabriel Kacso, MD](#) ^m · [Gwenaëlle Gravis, MD](#) ⁿ · [Fabio Calabro, MD](#) ^o · [Jean-François Berdah, MD](#) ^p · [Ali Hasbini, MD](#) ^q · [Marlon Silva, MD](#) ^r · [Antoine Thiery-Vuillemin, MD](#) ^s · [Igor Latorzeff, MD](#) ^t · [Loïc Mourey, MD](#) ^u · [Brigitte Laguerre, MD](#) ^v · [Sophie Abadie-Lacourtoisie, MD](#) ^w · [Etienne Martin, MD](#) ^x · [Claude El Kouri, MD](#) ^y · [Anne Escande, MD](#) ^z · [Alvar Rosello, MD](#) ^{aa} · [Nicolas Magne, MD](#) ^{ab} · [Friederike Schlurmann, MD](#) ^{ac} · [Frank Priou, MD](#) ^{ad} · [Marie-Eve Chand-Fouche, MD](#) ^{ae} · [Salvador Villà Freixa, MD](#) ^{af} · [Muhammad Jamaluddin, MD](#) ^{ag} · [Isabelle Rieger, MSc](#) ^{ah} · [Alberto Bossi, MD](#) ^d on behalf of the PEACE-1 investigators [†] [Show less](#)

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Publication History: Published April 8, 2022

DOI: [10.1016/S0140-6736\(22\)00367-1](https://doi.org/10.1016/S0140-6736(22)00367-1)Also available on [ScienceDirect](#)

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Summary

Background

Current standard of care for metastatic castration-sensitive prostate cancer supplements androgen deprivation therapy with either docetaxel, second-generation hormonal therapy, or radiotherapy. We aimed to evaluate the efficacy and safety of abiraterone plus prednisone, with or without radiotherapy, in addition to standard of care.

Methods

We conducted an open-label, randomised, phase 3 study with a 2×2 factorial design (PEACE-1) at 77 hospitals across Belgium, France, Ireland, Italy, Romania, Spain, and Switzerland. Eligible patients were male, aged 18 years or older, with histologically confirmed or cytologically confirmed de novo metastatic prostate adenocarcinoma, and an Eastern Cooperative Oncology Group performance status of 0–1 (or 2 due to bone pain). Participants were randomly assigned (1:1:1:1) to standard of care (androgen deprivation therapy alone or with intravenous docetaxel 75 mg/m² once every 3 weeks), standard of care plus radiotherapy, standard of care plus abiraterone (oral 1000 mg abiraterone once daily plus oral 5 mg prednisone twice daily), or standard of care plus radiotherapy plus abiraterone. Neither the investigators nor the patients were masked to treatment allocation. The coprimary endpoints were radiographic progression-free survival and overall survival. Abiraterone efficacy was first assessed in the overall population and then in the population who received androgen deprivation therapy with docetaxel as standard of care (population of interest). This study is ongoing and is registered with [ClinicalTrials.gov](https://clinicaltrials.gov), [NCT01957436](https://clinicaltrials.gov/ct2/show/study/NCT01957436).

Findings

Between Nov 27, 2013, and Dec 20, 2018, 1173 patients were enrolled (one patient subsequently withdrew consent for analysis of his data) and assigned to receive standard of care (n=296), standard of care plus radiotherapy (n=293), standard of care plus abiraterone (n=292), or standard of care plus radiotherapy plus abiraterone (n=291). Median follow-up was 3·5 years (IQR 2·8–4·6) for radiographic progression-free survival and 4·4 years (3·5–5·4) for overall survival. Adjusted Cox regression modelling revealed no interaction between abiraterone and radiotherapy, enabling the pooled analysis of abiraterone efficacy. In the overall population, patients assigned to receive abiraterone (n=583) had longer radiographic progression-free survival (hazard ratio [HR] 0·54, 99·9% CI 0·41–0·71; p<0·0001) and overall survival (0·82, 95·1% CI 0·69–0·98; p=0·030) than patients who did not receive abiraterone (n=589). In the androgen deprivation therapy with docetaxel population (n=355 in both with abiraterone and without abiraterone groups), the HRs were consistent (radiographic progression-free survival 0·50, 99·9% CI 0·34–0·71; p<0·0001; overall survival 0·75, 95·1% CI 0·59–0·95; p=0·017). In the androgen deprivation therapy with docetaxel population, grade 3 or worse adverse events occurred in 217 (63%) of 347 patients who received abiraterone and 181 (52%) of 350 who did not; hypertension had the largest difference in occurrence (76 [22%] patients and 45 [13%], respectively). Addition of abiraterone to androgen deprivation therapy plus docetaxel did not increase the rates of neutropenia, febrile neutropenia, fatigue, or neuropathy compared with androgen deprivation therapy plus docetaxel alone.

Interpretation

Combining androgen deprivation therapy, docetaxel, and abiraterone in de novo metastatic castration-sensitive prostate cancer improved overall survival and radiographic progression-free survival with a modest increase in toxicity, mostly hypertension. This triplet therapy could become a standard of care for these patients.

Funding

Janssen-Cilag, Ipsen, Sanofi, and the French Government.

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